



Optimal USA Gymnastics Lineup: Simulation for 2024 Olympics

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Background: Project Overview



Objective:

Select the optimal USA lineup for the 2024 Women's Gymnastics Olympic Games to maximize the medal performance.

Dataset Overview:

- Tokyo 2021 Olympic Results
- 2022-2023 International Competition Results Including 40+ major competitions, such as:
 - World Championships (2022, 2023)
 - European Championships (2022, 2023)

Approach:

Build a performance prediction model

- Predict each gymnast's score distribution based on historical data.

Fix the lineups of 11 other countries

Simulate all possible USA lineups (2000 simulations per lineup)

- Run the full team competition and individual events.
- Calculate each lineup's average medal count.

Select the optimal lineup

- Prioritize gold medals, then silver, then bronze.

Background: Paris 2024 Gymnastics Competition Format



Qualification Round ("543")

- 5 gymnasts per team
- 4 compete on each apparatus
- Top 3 scores count
- Goal: Qualify for team, all-around, and apparatus finals



Team Final ("533")

- Top 8 teams
- 3 gymnasts per apparatus
- All 3 scores count

Individual All-Around Final

- Top 24 gymnasts (max 2 per country)
- Compete on all 4 apparatuses
- Total score determines ranking

Apparatus Finals

- Top 8 gymnasts per apparatus (max 2 per country)
- Score on each apparatus(VT, UB, BB, and FX) determines ranking

Modeling Approach



country	round	apparatus	location	score	fullname	days_before_event	fullname_mean
CAN	qual	VT	Germany	11.45	Evandra Zlobec	494	12.112500
CAN	qual	UB	United Kingdom	10.55	Maya Zonneveld	723	10.550000
CHN	qual	BB	Singapore	13.30	Tong Zuo	403	13.466273

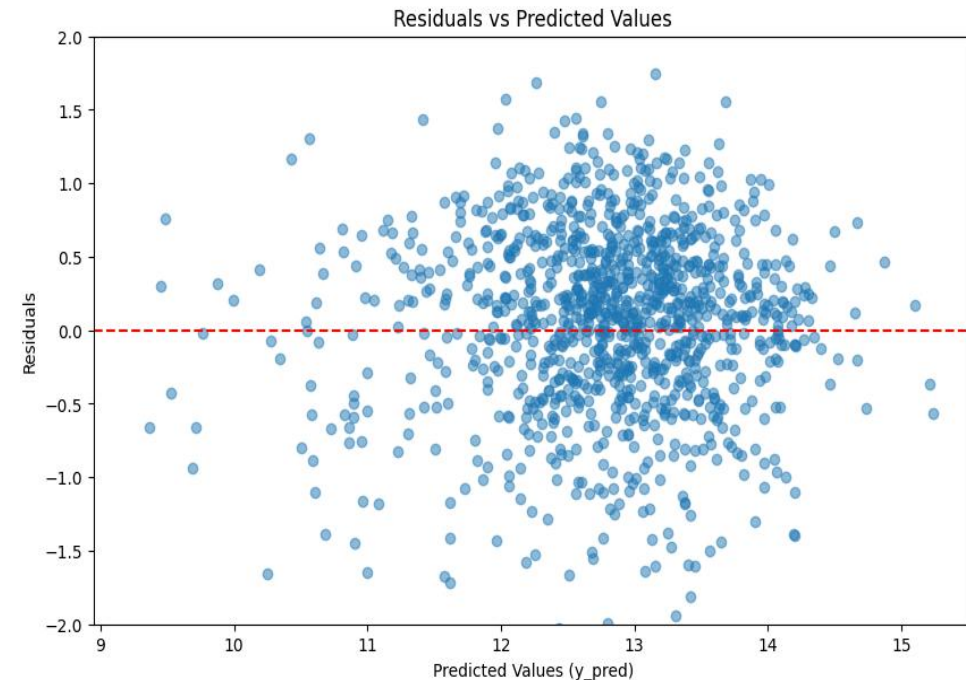
Feature Engineering

- Target Encoding
- One-hot Encoding
- Standardization

Models

- Linear Regression
- Random Forest
- Baseline Model

Linear Regression 5-Fold CV RMSE: 0.7273592320328124
Random Forest 5-Fold CV RMSE: 0.7618592213694821
Baseline Model (Mean Prediction) RMSE: 0.7600562291427706



Simulation Process



Step 1: Fix Lineups of 11 Countries

- Select top 5 gymnasts from each country based on historical average performance
- Fixed lineups: GBR, CAN, BRA, FRA, JPN, NED, ITA, AUS, ROU, KOR, CHN

Step 2: Predict Performance

- Predict scores for all U.S. gymnasts
- Predict scores for each gymnast in the fixed lineups
- Add random noise to mimic real competition uncertainty

Step 3: USA Lineup Simulation

- Enumerate all possible USA lineups
 - Simulate from qualification round (543 rule) to:
 - Team Final (533 rule)
 - Individual All-Around Final
 - Event Finals (VT, UB, BB, and FX)
 - Run 2000 simulations per lineup
 - Prioritize gold count
- (Use noise-free predicted scores to determine which athletes compete in team events.)

Simulation Result:

Best Team candidate (in order of gold medal priority):

	combination	overall_gold	overall_silver	overall_bronze	overall_total
(Simone Biles, Sunisa Lee, Jade Carey, Jordan Chiles, Leanne Wong)		3.1735	2.0455	1.6670	6.8860
(Simone Biles, Shilese Jones, Sunisa Lee, Jade Carey, Kayla Dicello)		3.1735	2.0285	1.6680	6.8700
(Simone Biles, Shilese Jones, Jade Carey, Jordan Chiles, Leanne Wong)		3.1710	2.0170	1.6575	6.8455
(Simone Biles, Shilese Jones, Sunisa Lee, Skye Blakely, Olivia Greaves)		3.1645	2.0345	1.6285	6.8275
(Simone Biles, Shilese Jones, Konnor McClain, Sunisa Lee, Kayla Dicello)		3.1645	1.9985	1.6645	6.8275

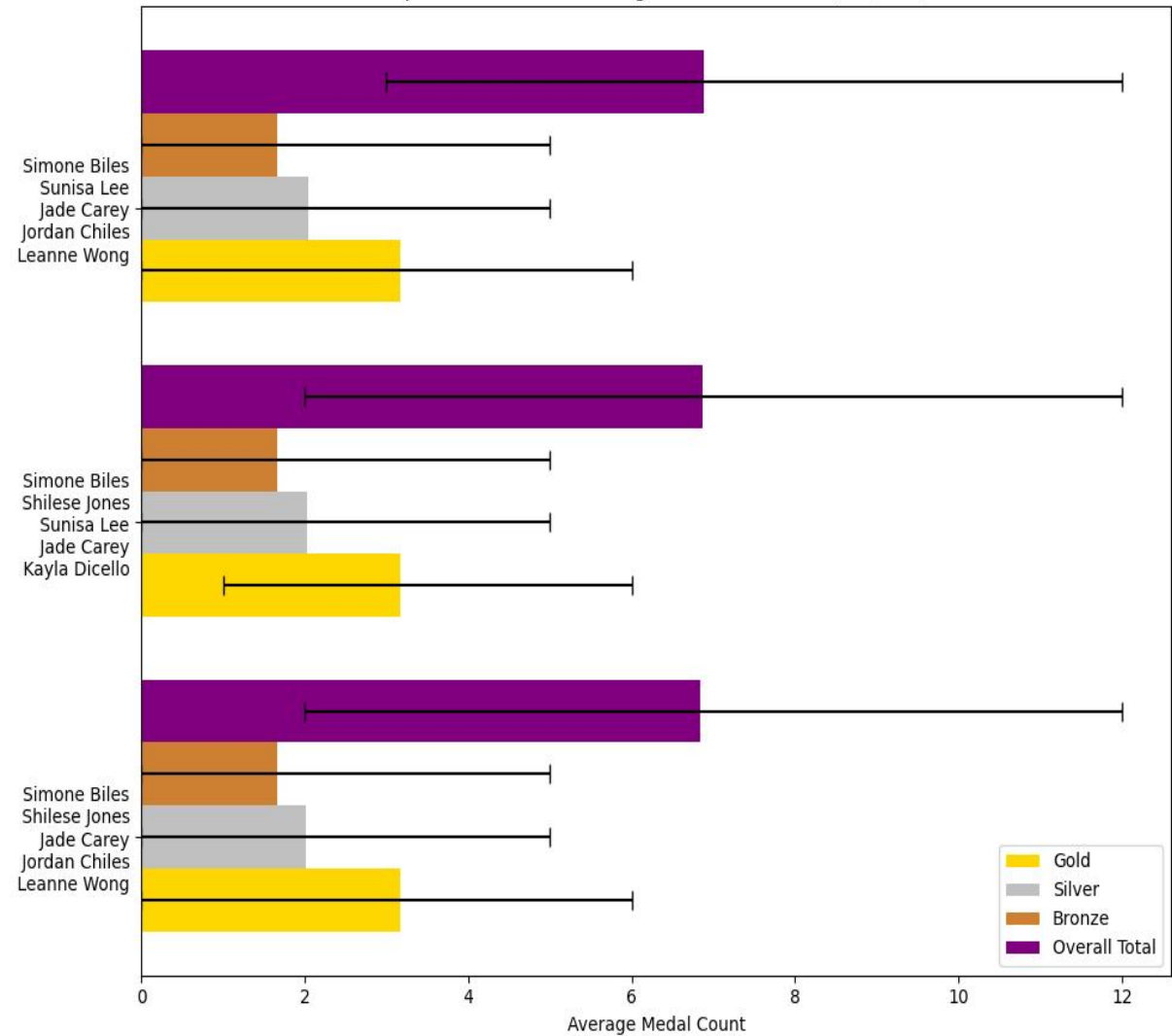
USA Lineup for 2024 Paris Olympics: Simone Biles; Sunisa Lee; Jade Carey; Jordan Chiles; Hezly Rivera
Traveling Replacement Alternates: Joscelyn Roberson; Leanne Wong

Results & Analysis

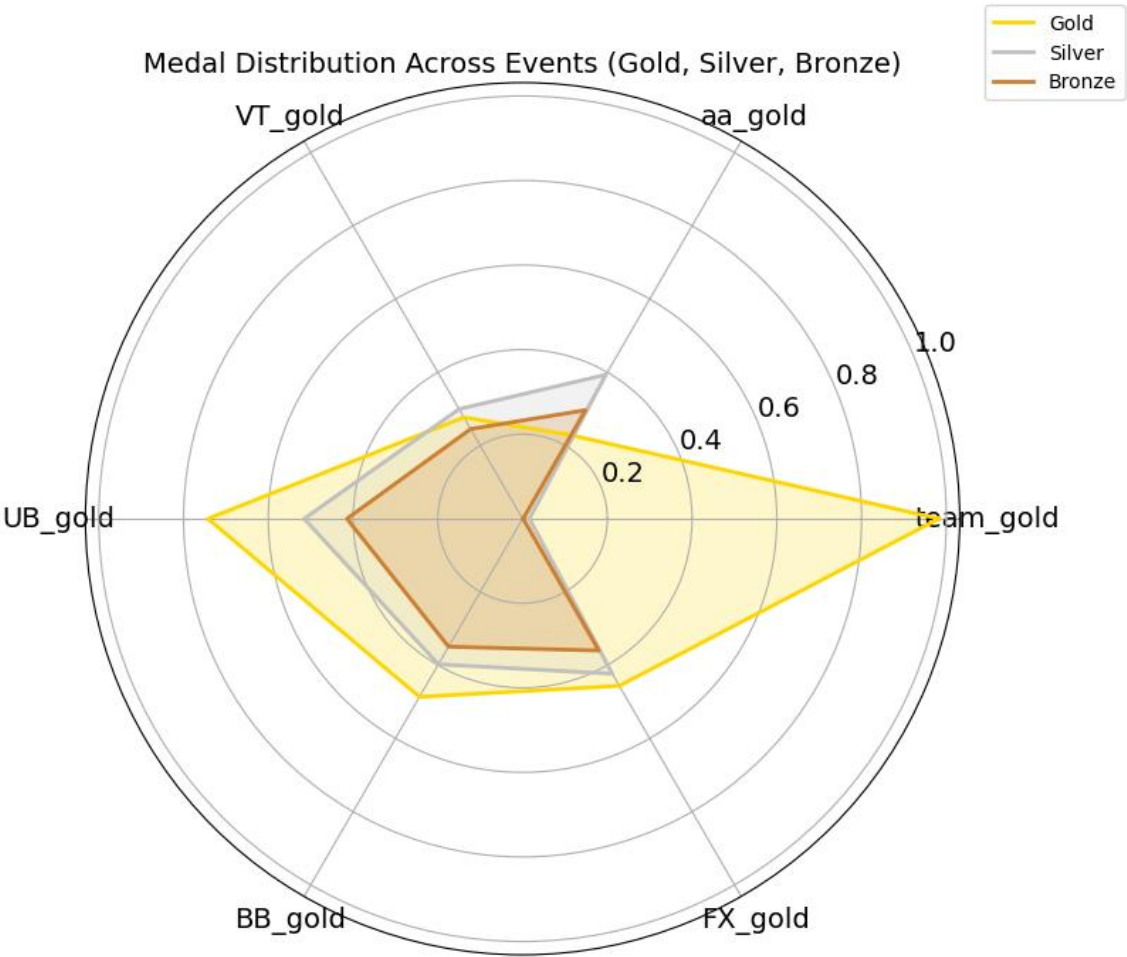


Overall Performance

Top 3 Teams Medal Averages with Error Bars (Min/Max)



Event-Specific Performance



Future Outlook: Potential Improvements & Applications



Potential Improvements

Tokyo 2021 Data Integration

- Attempted to merge Tokyo 2021 data with the 2022-2023 dataset. However, this led to decreased model accuracy, requiring further refinement.

Refined Handling of Scores

- Consider separating D-score, E-score, and Penalty for more precise analysis.

Data Gaps for Emerging Talents

- The dataset only covers up to 2023, missing rising stars for the 2024 Paris Olympics. Additional data collection is needed to improve prediction accuracy.

Applications

Broad Predictive Scope

- The model can be extended beyond U.S. women's gymnastics at the Paris Olympics to predict performance across different countries, genders, and competitions.

Comprehensive Performance Analysis

- Simulates optimal athlete selection for each event and identifies national strengths in specific apparatus. Also projects potential medal outcomes, especially gold medals.



Thank you

Q&A





Backup





	feature	coefficient	abs_coefficient
27	fullname_mean	0.958685	0.958685
7	apparatus_VT	0.467532	0.467532
4	apparatus_BB	-0.356576	0.356576
22	location_Slovenia	0.317358	0.317358
16	location_France	-0.260103	0.260103
19	location_Israel	-0.235973	0.235973
15	location_Egypt	-0.223087	0.223087
11	location_Chile	0.161565	0.161565
14	location_Croatia	-0.153761	0.153761
13	location_Colombia	0.151164	0.151164
21	location_Singapore	0.116389	0.116389
0	round_AAFinal	-0.105562	0.105562
2	round_final	0.100746	0.100746
5	apparatus_FX	-0.076174	0.076174
8	location_Azerbaijan	-0.066670	0.066670
9	location_Belgium	0.054247	0.054247
10	location_Bulgaria	0.039569	0.039569
25	location_United States	0.039472	0.039472
1	round_TeamFinal	0.038269	0.038269
18	location_Hungary	0.036943	0.036943
6	apparatus_UB	-0.034782	0.034782
3	round_qual	-0.033454	0.033454
24	location_United Kingdom	0.032517	0.032517
26	days_before_event	-0.021761	0.021761
20	location_Qatar	-0.004496	0.004496
17	location_Germany	-0.002417	0.002417
12	location_China	-0.001995	0.001995
23	location_Turkey	-0.000720	0.000720

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